

YellowCab Intelligence

NYC Yellow Taxi Trip Duration Predictor
Cleaned 46 Million Trips • Analyzed 2M Sample • 2025 Data
Powered by LightGBM • $R^2 = 0.86$

The Problem

NYC taxi passengers have no reliable way to know how long their trip will take before getting in.

Given: Pickup location, dropoff location, time of day

Predict: Trip duration in minutes

Why it matters: Better planning, smart dispatch, reduced passenger frustration

The Dataset

Source : NYC TLC Trip Record Data (Official Government Data)

URL: nyc.gov/site/tlc/about/tlc-trip-record-data.page

Period: January 2025 - December 2025 (Full Year)

Raw trips: 48,722,602

Clean trips after filtering: 46,573,205

Sample used for modeling: 2,000,000 trips

The Dataset (ii)

Key fields used:

- Pickup & dropoff location (263 NYC taxi zones)
- Trip distance
- Pickup datetime (hour, day, month)
- Fare amount
- Passenger count

The Dataset (iii)

Data cleaning removed:

- Trips under 1 minute or over 120 minutes
- Zero distance trips
- Missing location IDs

Key Insights from Exploration

Average trip duration: 14.5 minutes

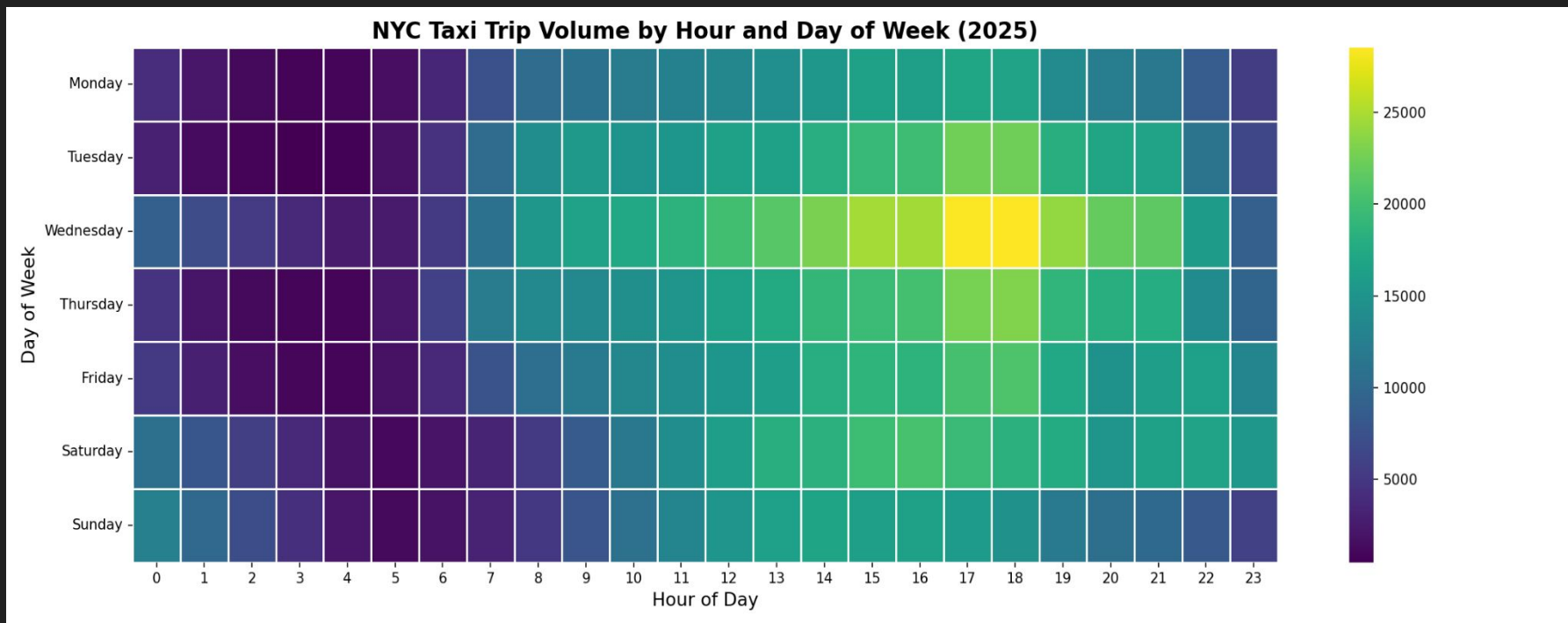
Rush hour trips (7-9am, 5-7pm) are significantly longer

Weekdays are busier than weekends

Top pickup zones:

- JFK Airport (# 1 busiest)
- Midtown Center
- Upper East Side

Heatmap shows Tuesday/Wednesday afternoons (3-5pm) are peak taxi demand periods



Feature Engineering (i)

9 smart features created to improve prediction:

Time Features:

- pickup_hour — hour of day
- day_of_week — Monday to Sunday
- month — seasonal patterns
- is_weekend — different travel behavior
- is_rush_hour — 7-9am and 5-7pm flag

Feature Engineering (ii)

Location Features:

- route_avg_duration — historical average for each pickup/dropoff pair
- congestion_fee_flag — automatically detected

from pickup location (not shown to user)

- pickup_borough
- dropoff_borough

Model Comparison (i)

4 models built and compared:

Model	RMSE	MAE	R ²
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Baseline	12.56	6.84	-0.26
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Linear Reg	5.29	3.56	0.78
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Ridge Reg	5.29	3.56	0.78
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Random Forest	4.22	2.65	0.86
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LightGBM	4.18	2.63	0.86
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Model Comparison (ii)

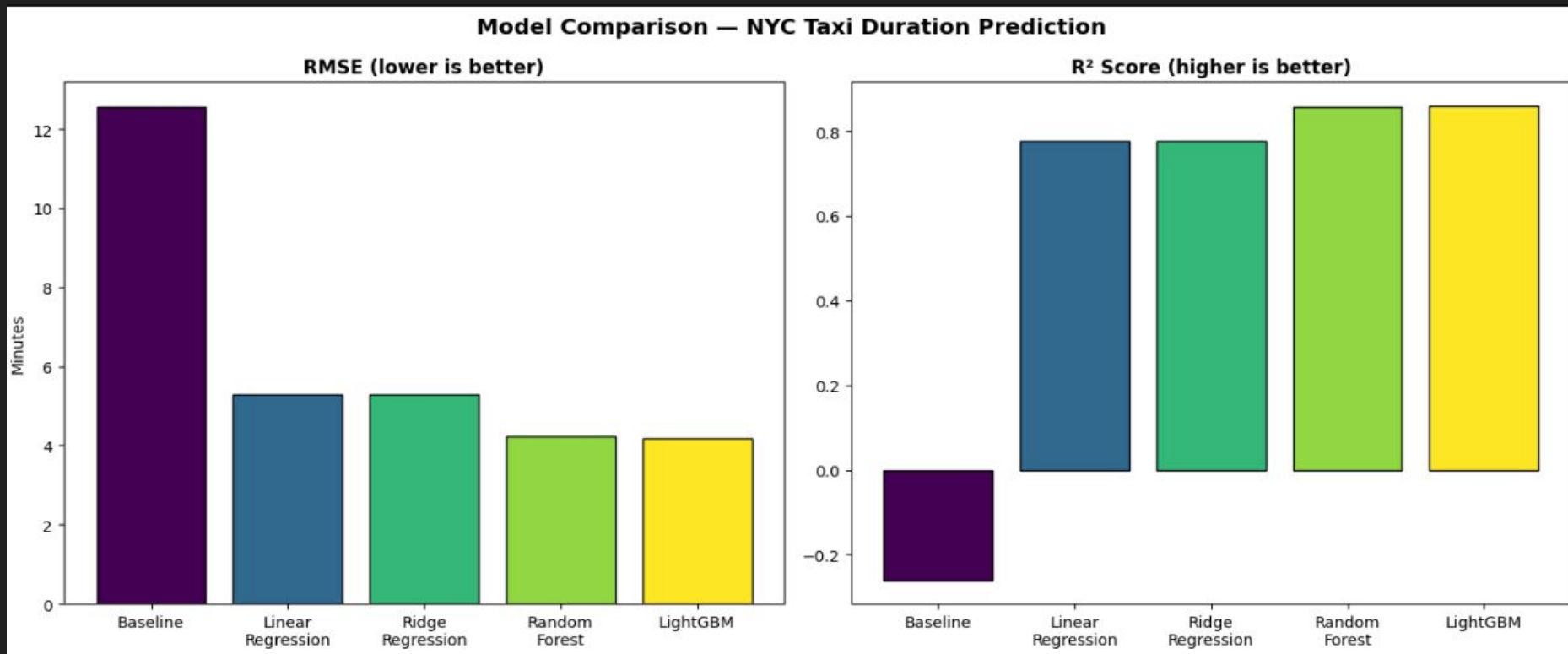
Winner: LightGBM

- Best RMSE: 4.18 minutes
- Best R^2 : 0.86
- Beats baseline by 67%!

All models used chronological train/test split

Train: 80% | Test: 20%

Model Comparison image



Live Streamlit App

Features:

- Select any NYC pickup & dropoff zone
- Choose hour of day and day of week
- Check rush hour flag
- Click Predict — get instant result!

Result displayed as:

"Estimated Trip Duration: X minutes"

Tech: Python • Streamlit • LightGBM

Runs locally: `python -m streamlit run app/streamlit_app.py`

Live Streamlit App (ii) image



NYC Yellow Taxi

Trip Duration Predictor

Powered by LightGBM • Trained on 46 Million NYC Trips



New York City • 2025 • Live Prediction Engine

Pickup Zone

Newark Airport (ID: 1)



Dropoff Zone

Newark Airport (ID: 1)



Hour of Day

Day of Week

Pickup Zone

Newark Airport (ID: 1)



Dropoff Zone

Newark Airport (ID: 1)



Hour of Day

8

Day of Week

Monday



Light Weight of Sample 5-7000



Predict My Trip Duration

Conclusion

What we achieved:

- Successfully predicted NYC taxi duration
- LightGBM achieved $R^2 = 0.86$
- Beats baseline model by 67%
- Built a live interactive prediction app
- Cleaned 46 million real NYC trips
- Analyzed 2 million trip sample for modeling

Conclusion

Key Findings:

- Pickup hour is the #1 predictor of duration
- Route history beats raw distance
- Tuesday/Wednesday afternoons are peak demand
- JFK Airport is NYC's busiest pickup zone

Future Improvements:

- Add real-time traffic data
- Include weather conditions